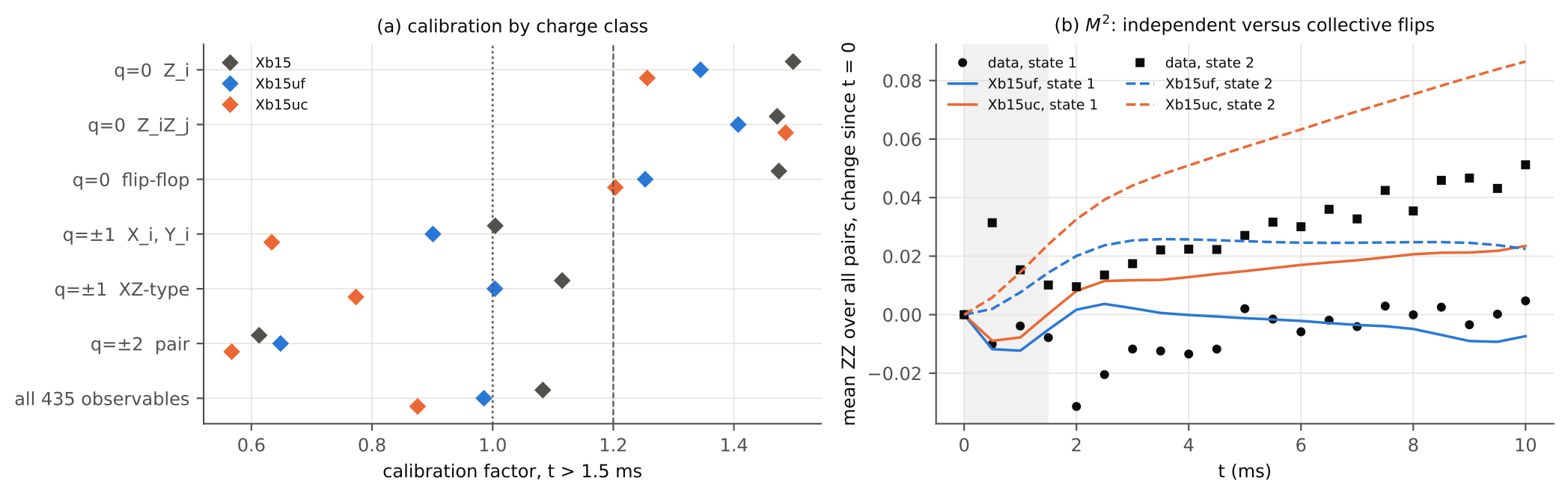
(a) calibration by charge class
Xb15
Xb15uf
Xb15uc
q=0 Z_i
q=0 Z_iZ_j
q=0 flip-flop
q=±1 X_i, Y_i
q=±1 XZ-type
q=±2 pair
all 435 observables
0.6
0.8
1.0
1.2
1.4
calibration factor, t > 1.5 ms
(b) M^2: independent versus collective flips
data, state 1
data, state 2
Xb15uf, state 1
Xb15uf, state 2
Xb15uc, state 1
Xb15uc, state 2
mean ZZ over all pairs, change since t = 0
0.08
0.06
0.04
0.02
0.00
−0.02
0
2
4
6
8
10
t (ms)